INCH-POUND

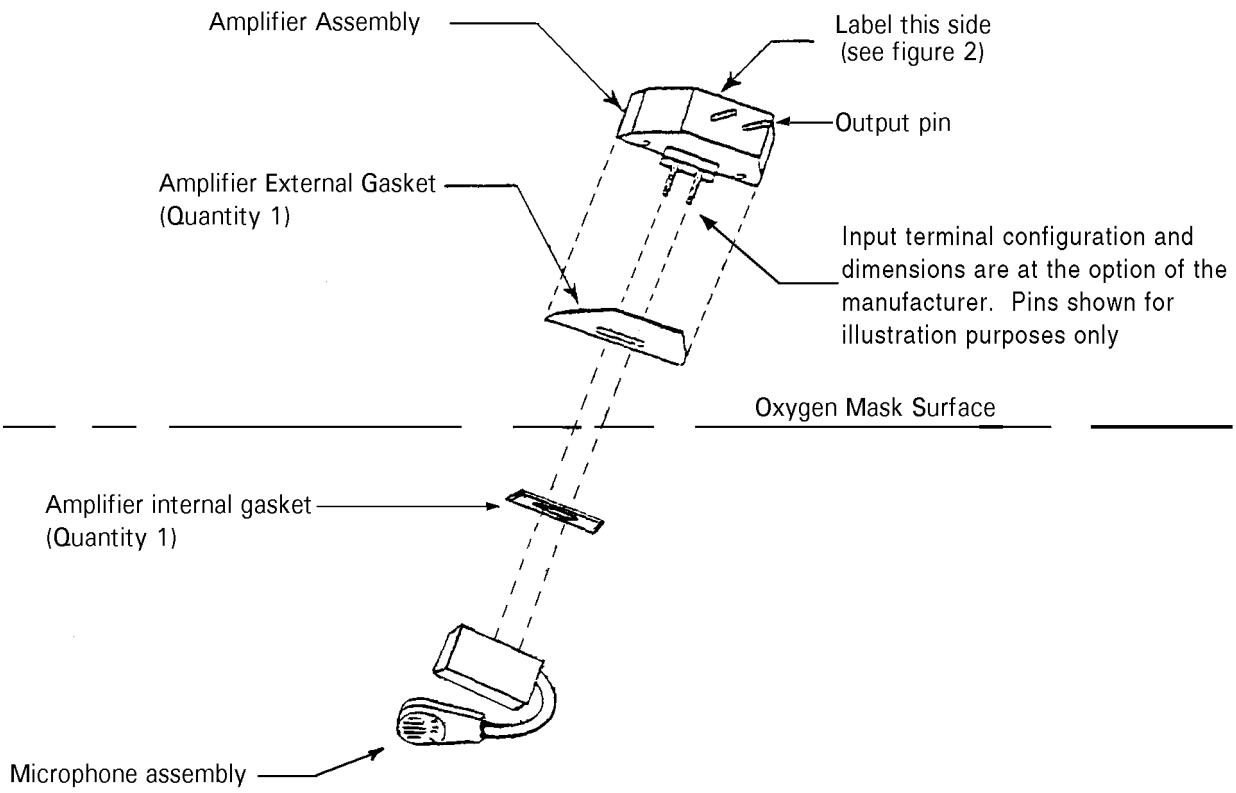
MIL-DTL-32015/2(AS)
15 October 1999

DETAIL SPECIFICATION SHEET

MICROPHONE, LINEAR, MBU-2 OXYGEN MASK M-176/AIC

This specification is approved for use by the Naval Air Systems Command, Department of Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-32015 listed in the issue of the Department of Defense Index of Specifications and Standards (DoDDIS) specified in the solicitation..



ANSC N/A FSC 5965

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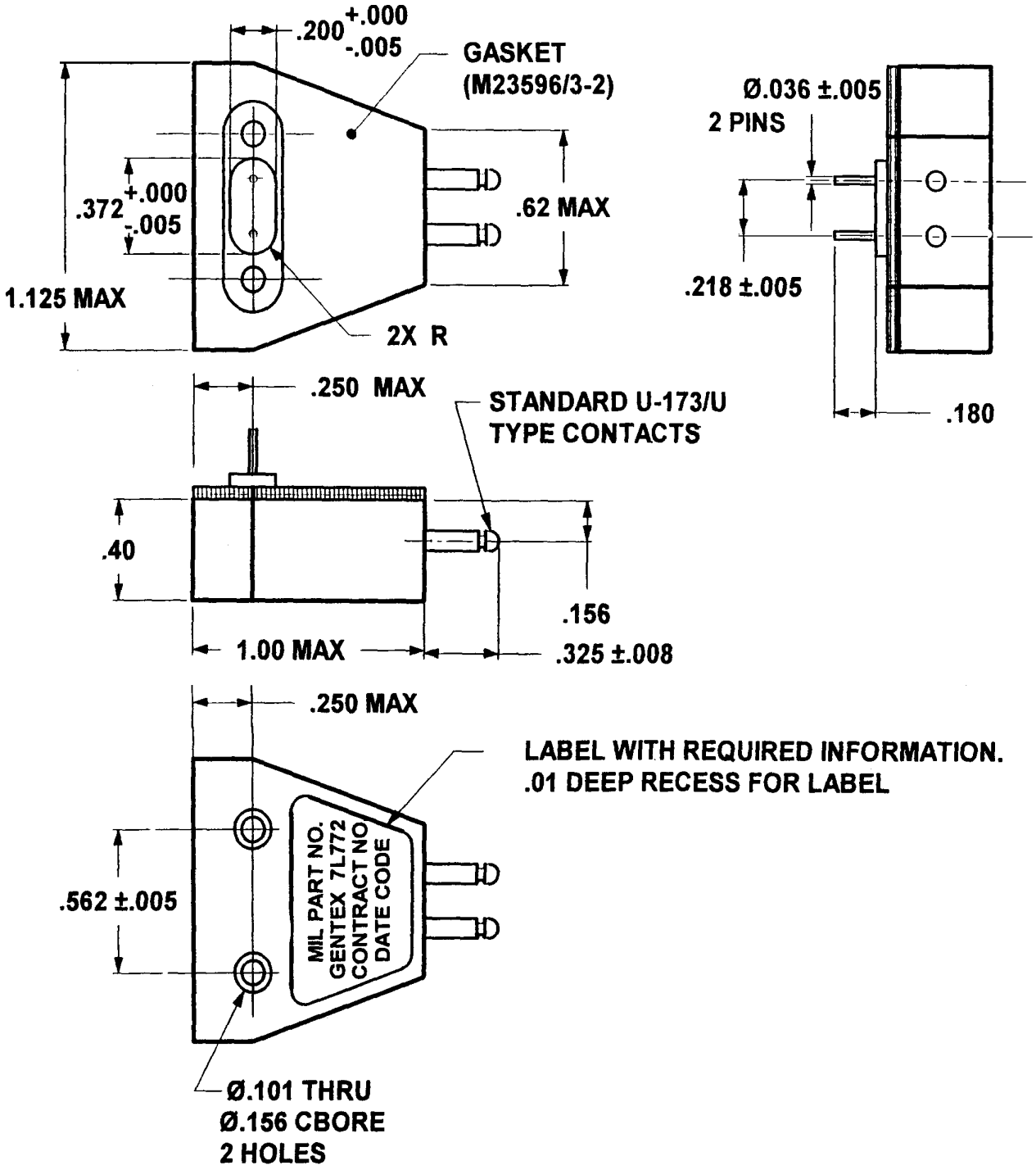


FIGURE 2. Amplifier assembly.

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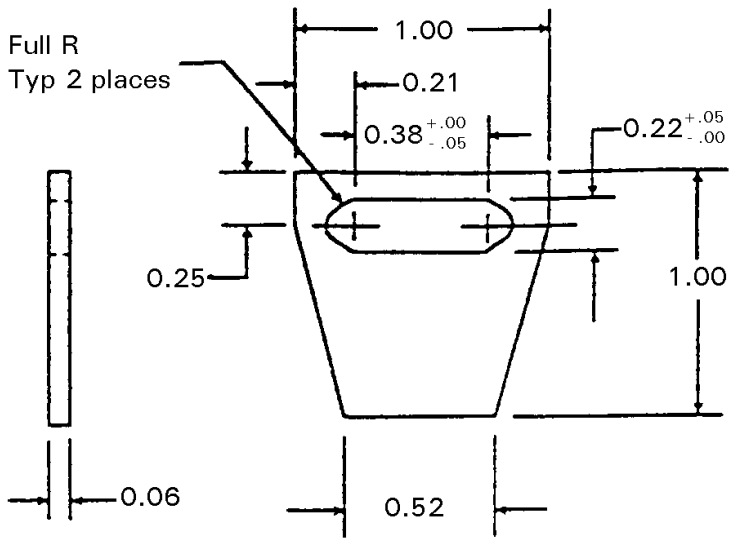


FIGURE 3a. Amplifier external gasket

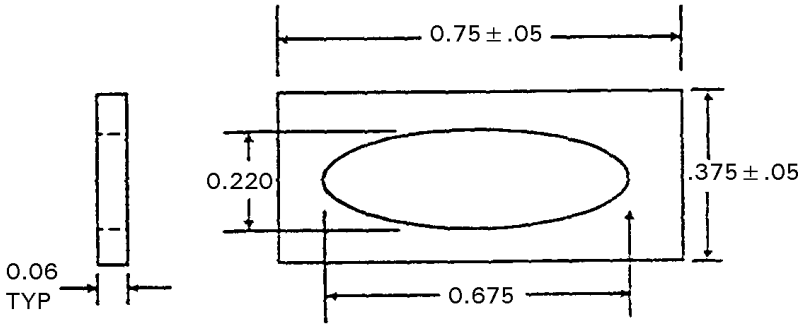


FIGURE 3b. Amplifier internal gasket.

NOTES:

1. All dimensions on Figure 3a are maximum except where indicated. Actual gasket dimensions shall be consistent with amplifier case design.
2. Material: Silicone Rubber 7 commercial grade.

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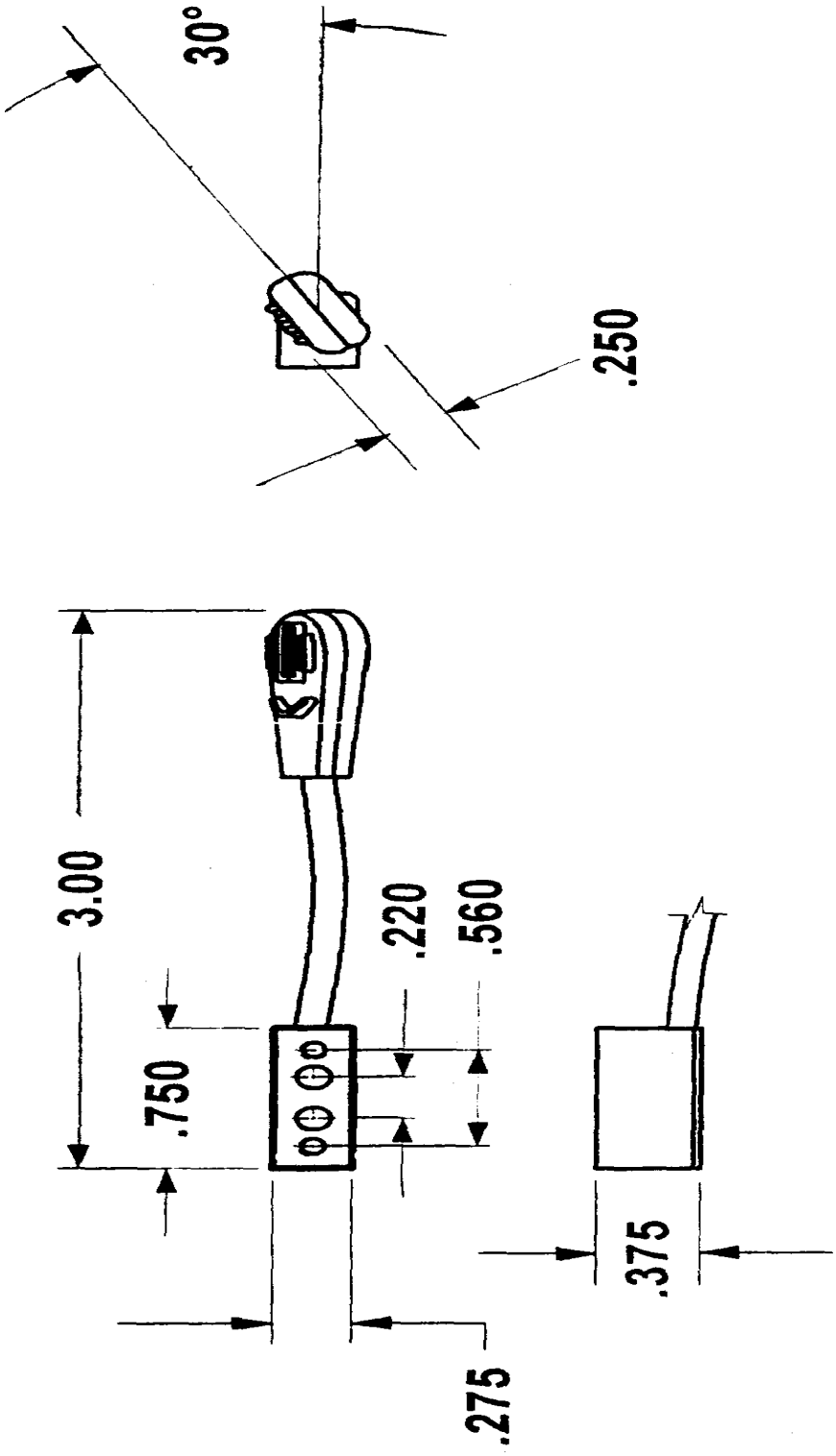


FIGURE 4. Microphone element and bracket.

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APPLICABLE DOCUMENTS:

MS35275	Screw, Machine-Drilled Fillister Head, Slotted, Corrosion Resisting Steel, passivated, UNC-2A

APPLICABLE DRAWINGS:

AIR FORCE

57B12662	Connector, Plug Electrical U-173/U
57B13664	Tip-Electrical, Plug Connector

REQUIREMENTS:

DESIGN and CONSTRUCTION: See figures1 through 4.

MICROPHONE/ AMPLIFIER INTERFACE: The microphone amplifier interface at the oxygen mask surface shall seal to protect the electrical contacts from severe moisture effects inside the oxygen mask. The internal gasket (see figure 3b) shall be permanently attached to the mounting bracket.

CONFIGURATION: The microphone assembly is comprised of a preamplifier assembly and a microphone element assembly. The microphone element assembly shall contain the microphone element, and only the electronics necessary for transfer of the audio signal to the preamplifier. To the maximum extent practical, all electronics shall be contained within the preamplifier assembly rather than the microphone element assembly to minimize the risk of a fire potential within the oxygen mask. Voltages and currents in the microphone element assembly shall not create sparks or any other adverse effect which presents a fire potential inside the oxygen mask.

COMPONENT PARTS: The microphone assembly shall be supplied with the following:

1 Amplifier assembly (see figure 2).
1 Gasket (see figure 3a) which shall be attached to the bottom of the amplifier via adhesive.
1 Gasket (see figure 3b).
1 Microphone element (see figure 4).
2 MS35275-210 machine screws, fillister head (2-56 x 3/4 lg.).

PART or IDENTIFYING NUMBERS(PIN): M32015/2-1

WEIGHT: 62g max. for the assembly.

COLOR: Semigloss, lusterless, black.

CONNECTOR: N/A

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OUTPUT PINS: The amplifier output terminals shall contain two pins made in accordance with Air Force Drawing 57B12664. The pins shall be arranged as detailed in Air Force Drawing 57B12662.

AMPLIFIER TO MICROPHONE PINS: The pin configuration is not specified since the pins are within the system design and not required to be compatible with external connectors. The connector design shall be within the envelope dimensions shown for compatibility with the oxygen mask design.

PIN STRESS. All amplifier pins shall be capable of a force of 45 pounds applied to the pins in a plane 90 degrees to their projection when tested in accordance with the pin stress test without cracking, splitting, separating or causing any other effect that would render the amplifier unserviceable.

PIN STRESS TEST. A snug-fitting pin block simulating the mating connector shall be attached to the pins and tested, as shown in Figure 8. The line of force shall extend 0.5 inch beyond the end of the pins opposite the amplifier body. The applied force rate shall be no less than 1 pound per second up to but not exceeding a force of 45 pounds. At the completion of the test, the amplifier shall meet the requirements of sensitivity to a close sound source, and frequency to a close sound source.

AMPLIFIER AND MICROPHONE BRACKET: The amplifier and microphone bracket shall be constructed so when assembled in the oxygen mask with the MS35275-210 screws they will not be damaged when 8 in lb. of torque is applied to the screws.

INPUT CURRENT: 17 mA max. at 27.5 VDC when connected as shown on figure 5 with switch SW1 open.

IMPEDANCE: The electrical impedance of the microphone assembly shall be such that the unit shall meet the sensitivity and frequency response to a close sound source while connected to a 100 ohm +1 percent load impedance.

SENSITIVITY: 96.4 mV minimum at 104 dB SPL with 27.5 VDC supply voltage, and shall not vary by more than 1.0 dB when the supply voltage is varied between 22 and 30 volts.

SENSITIVITY AT REDUCED BIAS VOLTAGE: Shall not vary by more than 12 dB at 104 dB SPL from the 27.5 VDC established value when the supply voltage is varied between 12 and 22 volts.

SENSITIVITY AT ALTITUDE: +3 dB of sensitivity at ground level while at a simulated altitude of 15,000 ft above sea level.

FREQUENCY RESPONSE AT GROUND LEVEL: See figure 6.

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FREQUENCY RESPONSE AT ALTITUDE: Within +5 dB of the frequency response at sea level while at a simulated altitude of 15,000 feet above sea level.

NOISE IMMUNITY: See figure 7.

LINEARITY: Resistive load shall be 100 +1 percent ohms.

HARMONIC DISTORTION: As specified in the general specification.

GAIN STABILIZATION: Within 100 ms when the supply voltage is between 23 and 30 VDC. Within 150 ms when supply voltage is between 12 and 22 VDC.

OVERVOLTAGE PROTECTION: 15 minutes operation in the test circuit shown on figure 5 with 27.5 volts applied while the 330 ohm resistor is shorted.

ENVIRONMENTAL STRESS SCREENING (ESS): 24 hours of temperature cycling between -20 and +85 ºC, with each cycle consisting of 1 hour at each temperature and the change between temperature occurring at a minimum rate of 5 ºC per minute. The microphone assemblies shall be functional tested prior to each cold to hot transition and upon completion of the temperature cycling.

MARKING:

M-176/AIC
M32015/2-01
MANUFACTURER'S NAME OR CAGE CODE
CONTRACT NUMBER
DATE CODE
SERIAL NUMBER

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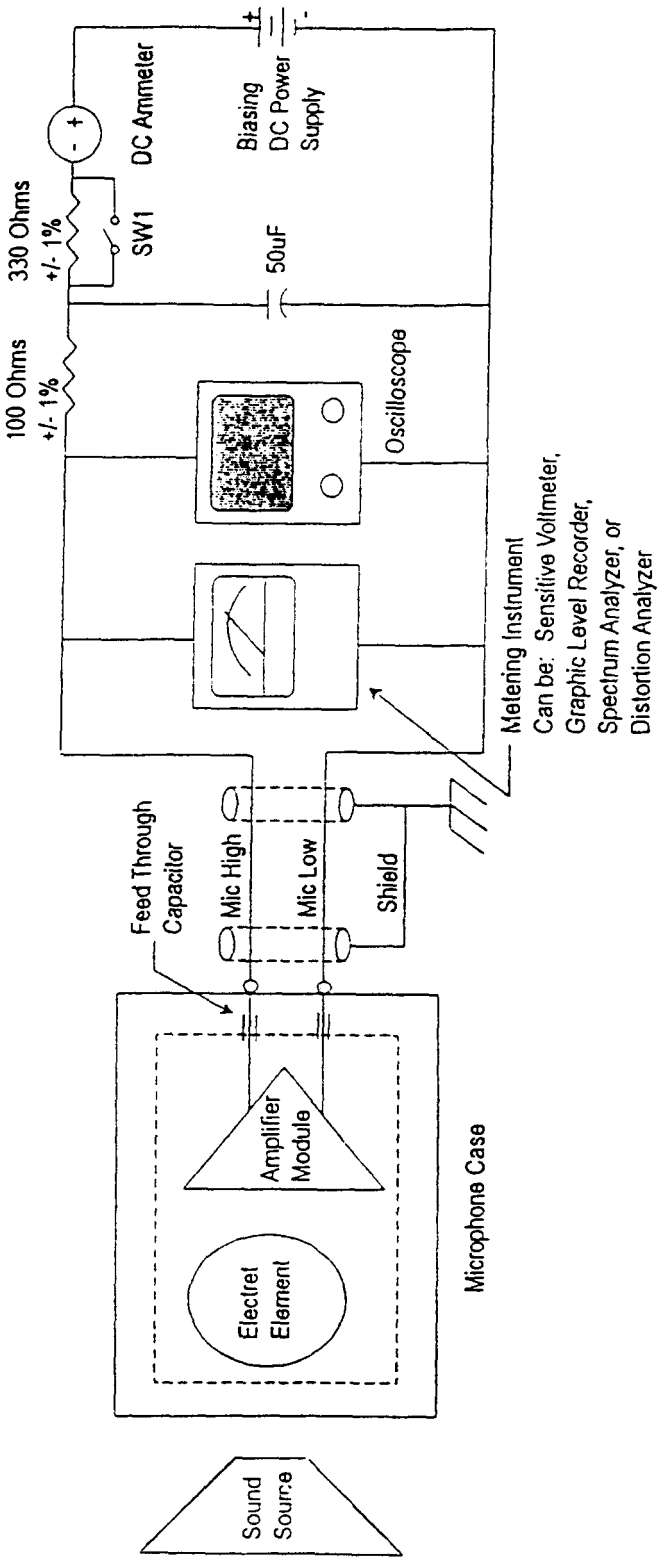


FIGURE 5. Test circuit schematic.

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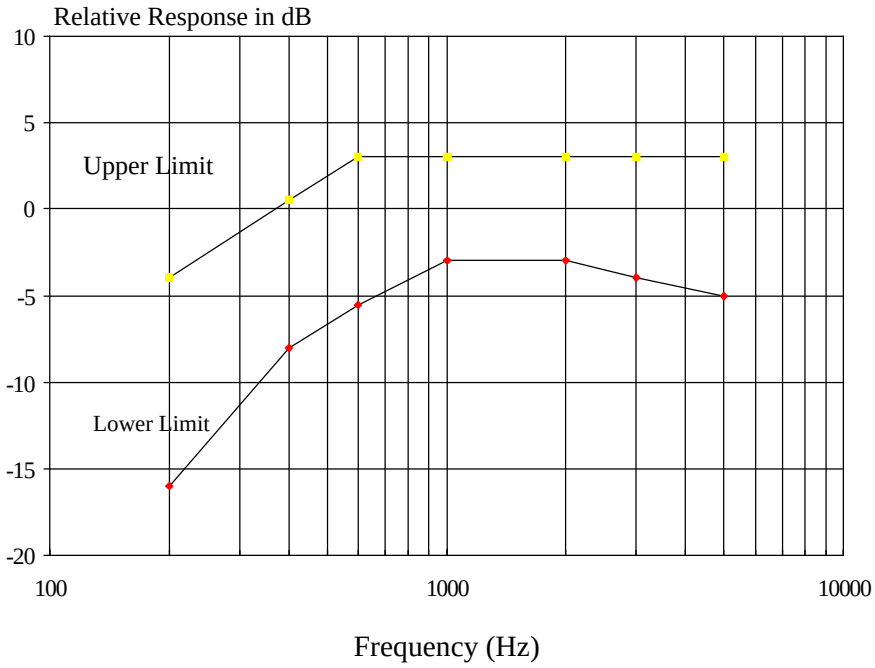


FIGURE 6 Frequency response limit curves

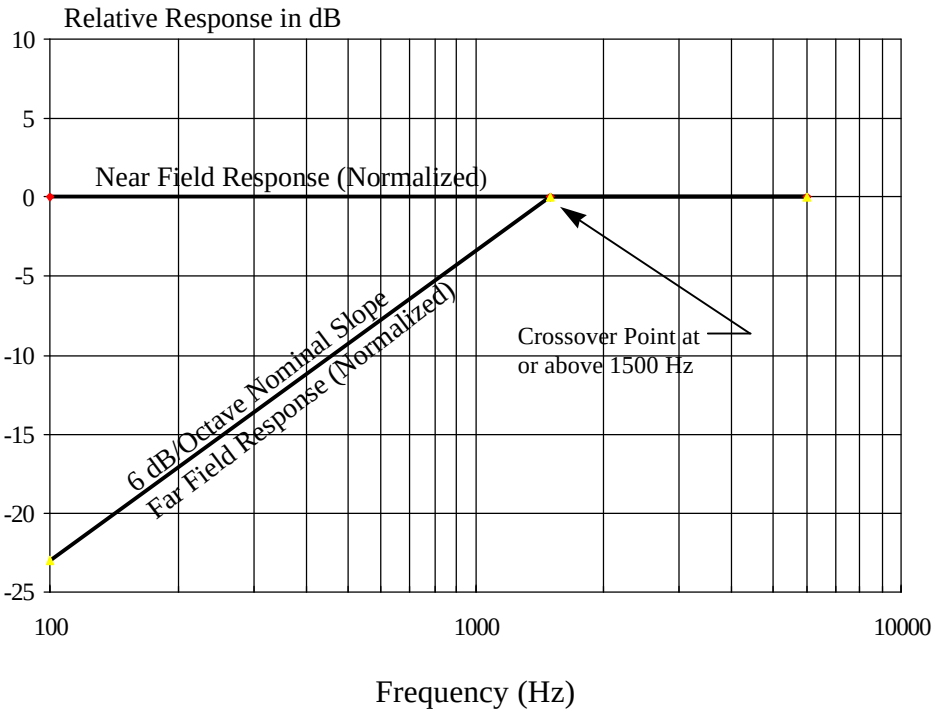
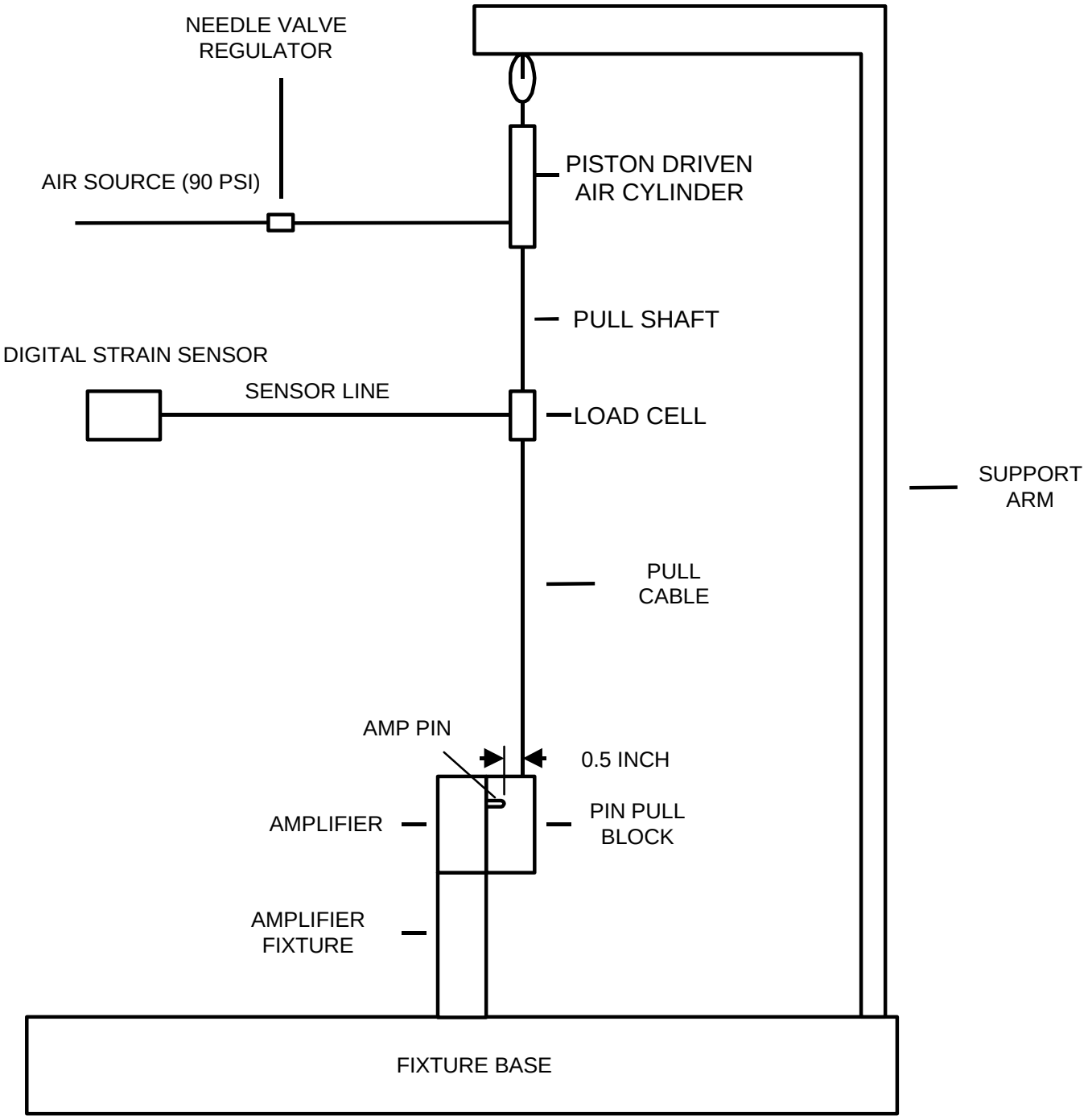


FIGURE 7. Noise immunity characteristics.

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NOTE:

This figure illustrates a typical test configuration only. Any equivalent test method is acceptable to achieve 45 pounds applied to the pins.

FIGURE 8. Typical amplifier pin stress to breakage test configuration.

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INTENDED USE: The microphone, M32015/2-1, is a noise-canceling, DC-powered, high-gain microphone designed for use in a pressure-type oxygen mask. The microphone is intended for use as a part of the airborne communication system which will provide communication of high intelligibility under the extreme noise conditions encountered in military aircraft. The microphone is designed to work with the AIC-14 intercommunication system, specifically control box C2645D.

Preparing activity:
Navy - As
(Project 5965-0269-02)

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